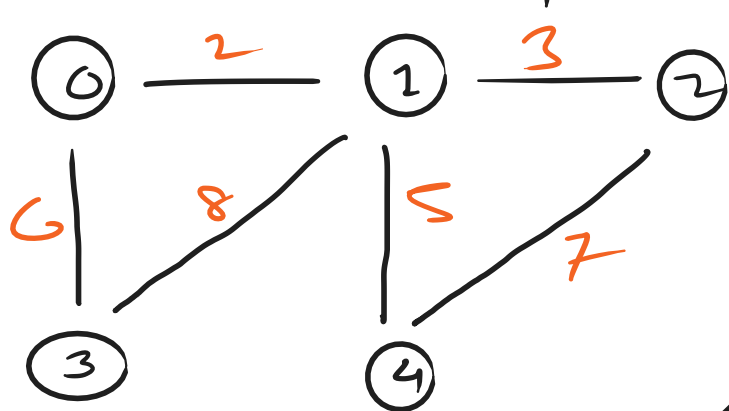


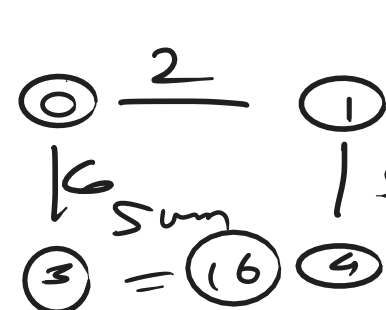
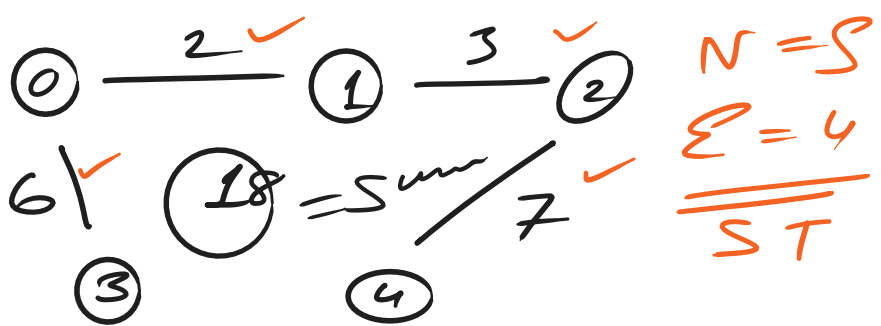
Minimum Spanning Tree : MST



$N \text{ nodes} = 5$
 $M \text{ Edges}$

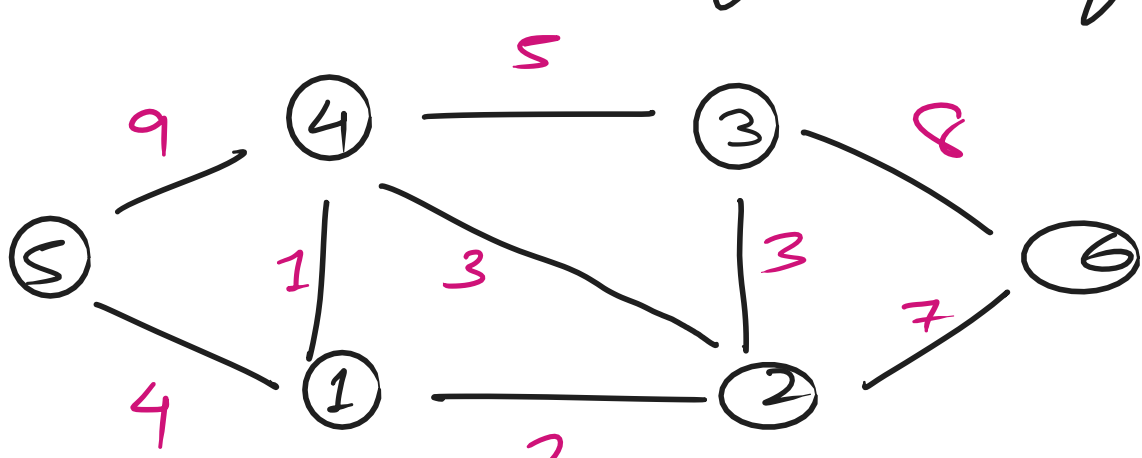
Spanning Tree

A tree in which we have N nodes and $N-1$ edges & "all nodes are reachable from all the other nodes"



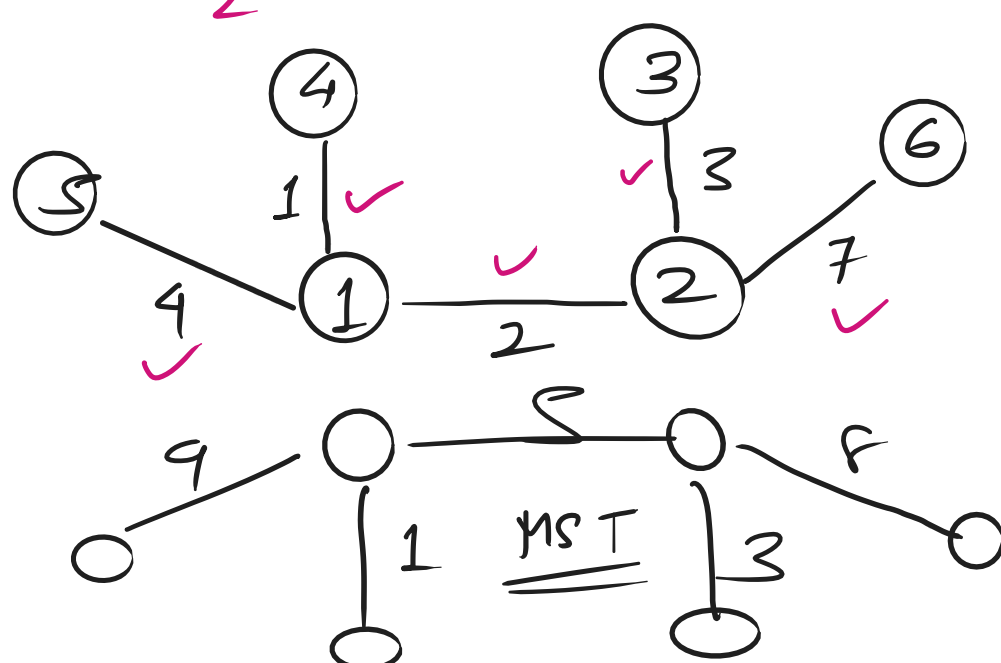
Tree having the least edge sum; weight sum. MST

* draw the MST for the following graph:



6 Nodes
MST = $(6-1) = 5$ edge

ST $\rightarrow$
MST $\rightarrow 1$



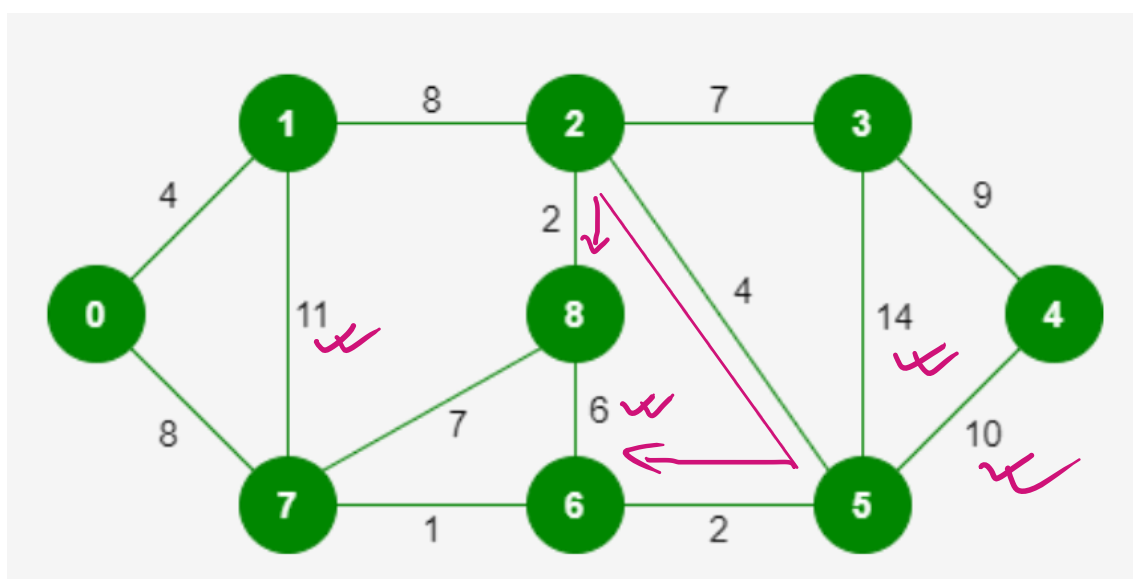
$7+3+4+2+1$
MST Sum = 17

H7-2
Try

F2F

MST

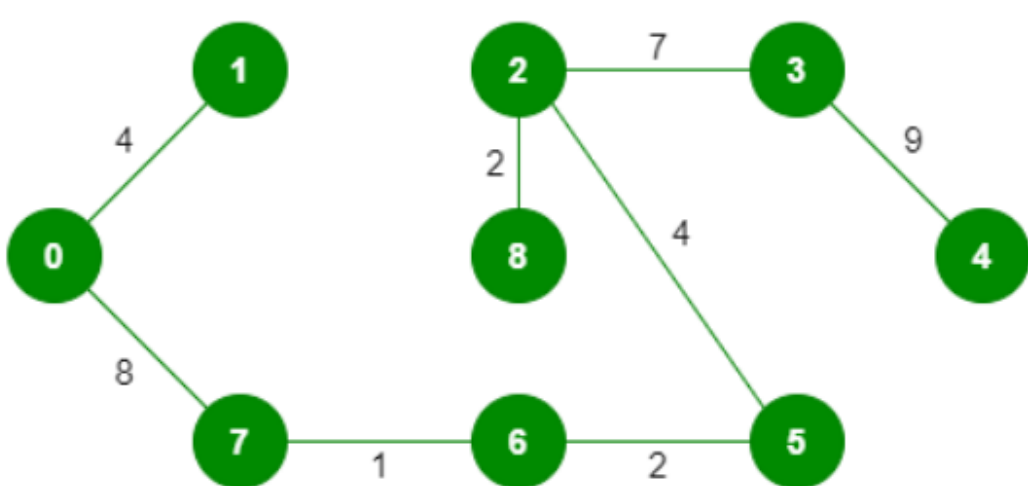
Algo



Prim's
Kruskal's

5 to 6
ST

MST

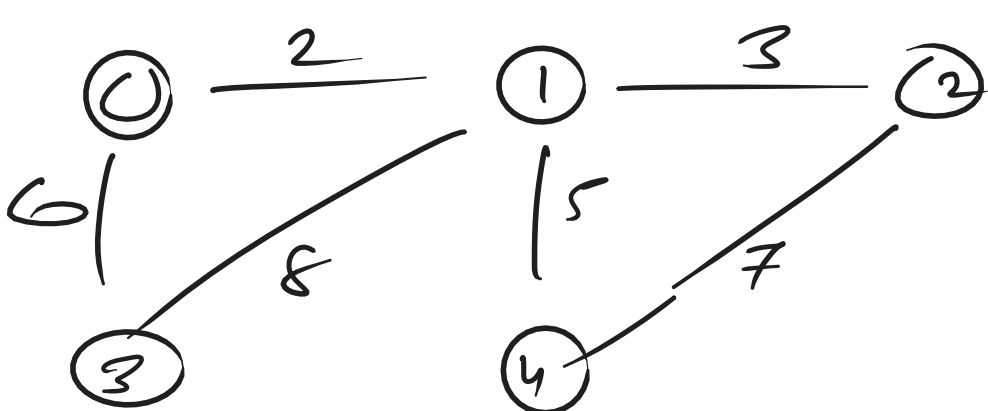


[Disjoint Set]

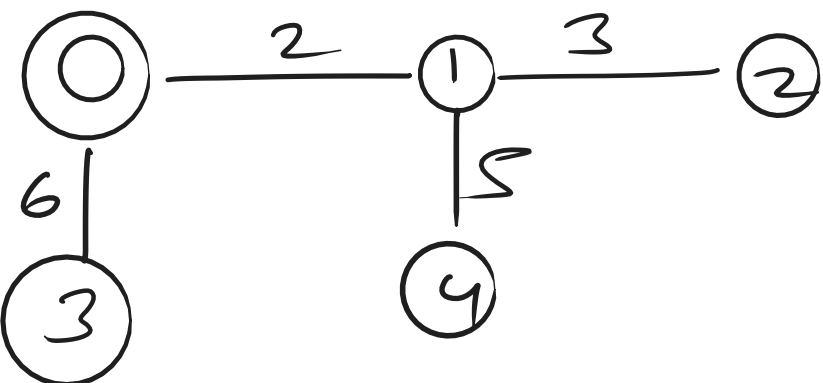
$\hookrightarrow$ Prim's
Kruskal's
 $\hookrightarrow$ Data Structure

Sum = 11

16



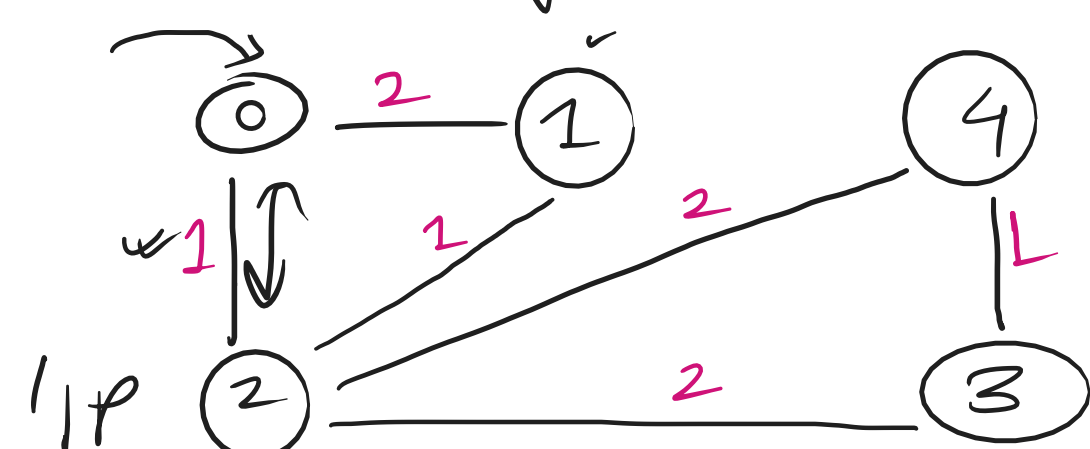
MST $\{(0,1), (1,2), (0,3), (1,4)\}$



Prim's Algorithm $\rightarrow$ Minimum Spanning Tree

[Sum, Edges]

Sum = $[0+1+1+2+1] = 5$



0	0, 0, -1
1	1, 2, 0
1	1, 1, 2
X	2, 1, 0
2	2, 3, 2
1	1, 4, 2

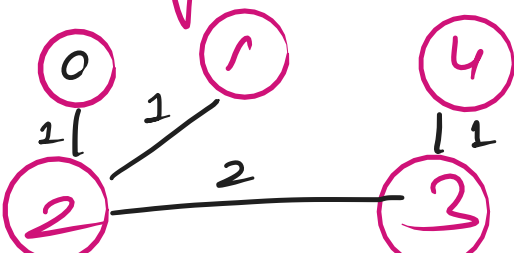
(2, 2, 3)
(1, 4, 3)
(2, 3, 2)
(2, 4, 2)
(1, 1, 2)
(1, 2, 0)
(2, 1, 0)
(0, 0, 1)

vis[] =

1	1	1	1	1
0	1	2	3	4

(wt, node, parent)
(min-heap)

Final MST Diagram:



MST Sum = 5

break $\rightarrow$ continue